

Bold Problem Homework 1

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Let $\|\cdot\|$ be the standard norm defined on $V = \mathbb{R}^n$ (Euclidean norm). That is,

$$\|x\| = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}$$

where $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

Definition (Inner Product). Let V be a vector space over a field F . An **inner product** on V is a function that assigns, to every ordered pair of vectors x and y in V , a scalar in F , denoted by $\langle x, y \rangle$ such that for all $x, y, z \in V$ and all $c \in F$ the following hold:

- (a) $\langle x + y, y \rangle = \langle x, y \rangle + \langle y, y \rangle$
- (b) $\langle cx, y \rangle = c\langle x, y \rangle$.
- (c) $\overline{\langle x, y \rangle} = \langle y, x \rangle$ where the bar denotes complex conjugation
- (d) $\langle x, x \rangle > 0$ if $x \neq 0$.

Lemma (Parallelogram Law). Let V be an inner product space. For all $x, y \in V$,

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

Proof. Let $x, y \in V$. Then we have

$$\begin{aligned} \|x + y\|^2 + \|x - y\|^2 &= \sum_{i=1}^n (x_i + y_i)^2 + \sum_{i=1}^n (x_i - y_i)^2 \\ &= \sum_{i=1}^n (x_i^2 + 2x_i y_i + y_i^2) + \sum_{i=1}^n (x_i^2 - 2x_i y_i + y_i^2) \\ &= 2 \sum_{i=1}^n x_i^2 + 2 \sum_{i=1}^n y_i^2 \\ &= 2\|x\|^2 + 2\|y\|^2. \end{aligned}$$

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Lemma. Let $\|\cdot\|$ be a norm on a real vector space V satisfying the Parallelogram law given in the lemma above. Then

$$\langle x, y \rangle = \frac{1}{4}[\|x + y\|^2 - \|x - y\|^2]$$

defines an inner product on V such that $\|x\|^2 = \langle x, x \rangle$ for all $x \in V$.

- (a) Prove $\langle x, 2y \rangle = 2\langle x, y \rangle$ for all $x, y \in V$.

Proof. Let $x, y \in V$. Then

$$\langle x, 2y \rangle = \frac{1}{4} [\|x + 2y\|^2 - \|x - 2y\|^2].$$

Observe that the expression inside the brackets of the equation above can be written as, using the parallelogram law,

$$\begin{aligned} \|x + 2y\|^2 - \|x - 2y\|^2 &= \|(x + y) + y\|^2 - \|(x - y) - y\|^2 \\ &= \|(x + y) + y\|^2 + \|(x + y) - y\|^2 \\ &\quad - (\|(x + y) - y\|^2 + \|(x - y) - y\|^2) \\ &= 2\|x + y\|^2 + 2\|y\|^2 - (2\|x - y\|^2 + 2\|y\|^2) \\ &= 2(\|x + y\|^2 - \|x - y\|^2). \end{aligned}$$

This tells us that

$$\langle x, 2y \rangle = 2 \cdot \frac{1}{4} [\|x + y\|^2 - \|x - y\|^2] = 2\langle x, y \rangle$$

which is our desired result. ■

(b) Prove $\langle x + u, y \rangle = \langle x, y \rangle + \langle u, y \rangle$ for all $x, u, y \in V$.

Proof. Let $x, y, u \in V$. Observe that the parallelogram law can be written as

$$\frac{1}{2} [\|x + y\|^2 + \|x - y\|^2] = \|x\|^2 + \|y\|^2.$$

Observe the sum

$$\langle x, y \rangle + \langle u, y \rangle = \frac{1}{4} [\|x + y\|^2 - \|x - y\|^2] + \frac{1}{4} [\|u + y\|^2 - \|u - y\|^2].$$

The first term can be written further as

$$\begin{aligned} \frac{1}{4} [\|x + y\|^2 - \|x - y\|^2] &= \frac{1}{4} \left[\frac{1}{2} [\|(x + y) + u\|^2 + \|(x + y) - u\|^2] \right. \\ &\quad \left. - \frac{1}{2} [\|(x - y) + u\|^2 + \|(x - y) - u\|^2] \right] \\ &= \frac{1}{4} \left[\frac{1}{2} [\|(x + u) + y\|^2 + \|(x + y) - u\|^2] \right. \\ &\quad \left. - \frac{1}{2} [\|(x + u) - y\|^2 + \|(x - y) - u\|^2] \right]. \end{aligned}$$

Likewise, the second term can be written as

$$\begin{aligned} \frac{1}{4} [\|u + y\|^2 - \|u - y\|^2] &= \frac{1}{4} \left[\frac{1}{2} [\|(u + y) + x\|^2 + \|(u + y) - x\|^2] \right. \\ &\quad \left. - \frac{1}{2} [\|(u - y) + x\|^2 + \|(u - y) - x\|^2] \right] \\ &= \frac{1}{4} \left[\frac{1}{2} [\|(u + x) + y\|^2 + \|(x - y) - u\|^2] \right. \\ &\quad \left. - \frac{1}{2} [\|(x + u) - y\|^2 + \|(x + y) - u\|^2] \right]. \end{aligned}$$

As a whole, we have

$$\begin{aligned} \langle x, y \rangle + \langle u, y \rangle &= \frac{1}{4} [\|(x + u) + y\|^2 - \|(x + u) - y\|^2] \\ &= \langle x + u, y \rangle. \end{aligned}$$
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- (c) Prove $\langle nx, y \rangle = n\langle x, y \rangle$ for every positive integer n and every $x, y \in V$.

Proof. Let $x, y \in V$. Let us proceed via induction to show the result. Note that the base case is trivial; that is, if $n = 1$, then we have $\langle x, y \rangle = \langle x, y \rangle$. Suppose that the result holds for all positive integers less than n . Then observe that $\langle (n-1)x, y \rangle = (n-1)\langle x, y \rangle$. Using part (b), we can write

$$\begin{aligned}\langle nx, y \rangle &= \langle ((n-1) + 1)x, y \rangle = \langle (n-1)x, y \rangle + \langle x, y \rangle \\ &= (n-1)\langle x, y \rangle + \langle x, y \rangle \\ &= ((n-1) + 1)\langle x, y \rangle \\ &= n\langle x, y \rangle.\end{aligned}$$

Thus, we see that $\langle nx, y \rangle = n\langle x, y \rangle$ which is our desired result. ■

- (d) Prove $m\langle \frac{1}{m}x, y \rangle = \langle x, y \rangle$ for every positive integer m and every $x, y \in V$.

Proof. Let $x, y \in V$ and let m be any positive integer. Then set $z = \frac{1}{m}x$. By using part (c), we can write

$$\begin{aligned}m\left\langle \frac{1}{m}x, y \right\rangle &= m\langle z, y \rangle \\ &= \langle mz, y \rangle \\ &= \left\langle m \cdot \frac{1}{m}x, y \right\rangle = \langle x, y \rangle.\end{aligned}$$

Thus, we have

$$m\left\langle \frac{1}{m}x, y \right\rangle = \langle x, y \rangle.$$

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- (e) Prove $\langle rx, y \rangle = r\langle x, y \rangle$ for every rational number r and every $x, y \in V$.

Proof. Let $x, y \in V$ and let $r \in \mathbb{Q}$. Observe that, by definition of rational numbers, we have $r = p/q$ with $p, q \in \mathbb{Z}$ and $q \neq 0$. Then we write

$$\begin{aligned}\langle rx, y \rangle &= \left\langle \frac{p}{q} \cdot x, y \right\rangle \\ &= p\left\langle \frac{1}{q}x, y \right\rangle && \text{(Part (c))} \\ &= \frac{p}{q}\langle x, y \rangle && \text{(Part (d))} \\ &= r\langle x, y \rangle.\end{aligned}$$

Thus, we have that $\langle rx, y \rangle = r\langle x, y \rangle$. ■

- (f) Prove $|\langle x, y \rangle| \leq \|x\|\|y\|$ for every $x, y \in V$.

Proof. Let $x, y \in V$. Using the triangle inequality for norms, we must have

$$\begin{aligned}|\langle x, y \rangle| &= \frac{1}{4} \left[\|x + y\|^2 - \|x - y\|^2 \right] \\ &\leq \frac{1}{4} \left[(\|x\| + \|y\|)^2 - \|x - y\|^2 \right] \\ &= \frac{1}{4} \left[2\|x\|\|y\| - (\|x - y\|^2 - \|x\|^2 - \|y\|^2) \right] \\ &\leq \|x\|\|y\|.\end{aligned}$$

Thus, we have that $|\langle x, y \rangle| \leq \|x\|\|y\|$. ■

- (g) Prove that for every $c \in \mathbb{R}$, every rational number r , and every $x, y \in V$,

$$|c\langle x, y \rangle - \langle cx, y \rangle| = |(c-r)\langle x, y \rangle - \langle (c-r)x, y \rangle| \leq 2|c-r|\|x\|\|y\|.$$

Proof. Let $c \in \mathbb{R}$, $r \in \mathbb{Q}$, and $x, y \in V$. Using part (e), we can write

$$\begin{aligned} |c\langle x, y \rangle - \langle cx, y \rangle| &= |c\langle x, y \rangle - r\langle x, y \rangle + r\langle x, y \rangle - \langle cx, y \rangle| \\ &= |(c-r)\langle x, y \rangle + \langle (r-c)x, y \rangle| \\ &= |(c-r)\langle x, y \rangle - \langle (c-r)x, y \rangle|. \end{aligned}$$

Working with the second equality, we can use the triangle inequality to write

$$\begin{aligned} |(c-r)\langle x, y \rangle + \langle (r-c)x, y \rangle| &\leq |(c-r)\langle x, y \rangle| + |\langle (r-c)x, y \rangle| \\ &= |c-r||\langle x, y \rangle| + |\langle (r-c)x, y \rangle| \\ &\leq |c-r|\|x\|\|y\| + \|(r-c)x\|\|y\| \\ &= |c-r|\|x\|\|y\| + |c-r|\|x\|\|y\| \\ &= 2|c-r|\|x\|\|y\|. \end{aligned}$$

Hence, we conclude that

$$|c\langle x, y \rangle - \langle cx, y \rangle| = |(c-r)\langle x, y \rangle - \langle (c-r)x, y \rangle| \leq 2|c-r|\|x\|\|y\|.$$

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- (h) Use the fact that for any $c \in \mathbb{R}$, $|c-r|$ can be made arbitrarily small, where r varies over the set of rational numbers, to establish item (b) of the definition of inner product.

Proof. Let $x, y \in V$ be non-zero. Let $c \in \mathbb{R}$ and let $\varepsilon > 0$. Suppose that

$$|c-r| < \frac{\varepsilon}{2\|x\|\|y\|}.$$

Then we have

$$|c\langle x, y \rangle - \langle cx, y \rangle| \leq 2|c-r|\|x\|\|y\| < \varepsilon$$

which is our desired result. ■

Proof of $\langle \cdot, \cdot \rangle$ is an inner product

Proof. Note that property (b) from above implies linearity in the first component of the inner product and property (g) implies that $c\langle x, y \rangle = \langle cx, y \rangle$ for all $c \in \mathbb{R}$ implies the scaling property of the first component is satisfied. Also, $\langle x, x \rangle > 0$ for all $x \in \mathbb{R}^n$ because

$$\langle x, x \rangle = \frac{1}{4} \left(\|2x\|^2 - \|x-x\|^2 \right) = \|x\|^2 > 0.$$

Also, $\langle x, y \rangle = \overline{\langle x, y \rangle}$ because $\langle x, y \rangle$ is a real number. ■

Problem 1. Suppose $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an isometry and $h(0) = 0$. Then h preserves distances and inner products on $\mathbb{R}^n$.

Proof. Since $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an isometry, we see that distance in $\mathbb{R}^n$ is preserved; that is, the distance function $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$d(x, y) = \|x - y\|$$

for all $x, y \in \mathbb{R}^n$ where $\|\cdot\|$ is a norm on defined on $\mathbb{R}^n$ and satisfies the following property:

$$d(h(x), h(y)) = d(x, y).$$

In particular, in terms of the $\|\cdot\|$ on $\mathbb{R}^n$, we have for all $x, y \in \mathbb{R}^n$

$$\|h(x) - h(y)\| = d(h(x), h(y)) = d(x, y) = \|x - y\|.$$

Since we can recover the norm (proven above) $\langle \cdot, \cdot \rangle$ defined by

$$\langle x, y \rangle = \frac{1}{4} \left(\|x + y\|^2 - \|x - y\|^2 \right).$$

Then h also preserves this inner product because for all $x, y \in \mathbb{R}^n$, we have

$$\begin{aligned}
 \langle h(x), h(y) \rangle &= \frac{1}{4} \left(\|h(x) + h(y)\|^2 - \|h(x) - h(y)\|^2 \right) \\
 &= \frac{1}{4} \left(2\|h(x)\|^2 + 2\|h(y)\|^2 - 2\|h(x) - h(y)\|^2 \right) \\
 &= \frac{1}{4} (2\|x\|^2 + 2\|y\|^2 - 2\|x - y\|^2) \\
 &= \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2) \\
 &= \langle x, y \rangle
 \end{aligned}$$

and we are done. ■

Problem 2. Suppose $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an isometry and $h(0) = 0$. Prove that h is necessarily a linear transformation.

Proof. Our goal is to show that h is a linear transformation; that is, for all $x, y \in \mathbb{R}^n$, we have

$$h(x + y) = h(x) + h(y)$$

and for all $\alpha \in \mathbb{R}$, we have

$$h(\alpha x) = \alpha h(x).$$

It suffices to show that for all $x, y \in \mathbb{R}^n$

$$\|h(x + y) - h(x) - h(y)\| = 0 \tag{1}$$

and for all $\alpha \in \mathbb{R}$,

$$\|h(\alpha x) - \alpha h(x)\| = 0. \tag{2}$$

First, we prove (1). Let $x, y \in \mathbb{R}^n$ be arbitrary. We have

$$\begin{aligned}
 \|h(x + y) - h(x) - h(y)\|^2 &= \langle h(x + y) - h(x) - h(y), h(x + y) - h(x) - h(y) \rangle \\
 &= \langle h(x + y), h(x + y) - h(x) - h(y) \rangle - \langle h(x), h(x + y) - h(x) - h(y) \rangle \\
 &\quad - \langle h(y), h(x + y) - h(x) - h(y) \rangle \\
 &= \langle h(x + y), h(x + y) \rangle - \langle h(x + y), h(x) \rangle - \langle h(x + y), h(y) \rangle \\
 &\quad - \langle h(x), h(x + y) \rangle + \langle h(x), h(x) \rangle + \langle h(x), h(y) \rangle - \langle h(y), h(x + y) \rangle + \langle h(y), h(x) \rangle + \langle h(y), h(y) \rangle \\
 &= \langle x + y, x + y \rangle - \langle x + y, x \rangle - \langle x + y, y \rangle - \langle x, x + y \rangle + \langle x, x \rangle + \langle x, y \rangle - \langle y, x + y \rangle + \langle y, x \rangle + \langle y, y \rangle \\
 &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle - \langle x, x \rangle - \langle y, x \rangle - \langle x, y \rangle - \langle y, y \rangle - \langle x, x \rangle - \langle x, y \rangle + \langle x, x \rangle + \langle x, y \rangle \\
 &\quad - \langle y, x \rangle - \langle y, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\
 &= 0.
 \end{aligned}$$

Hence, we have

$$\|h(x + y) - h(x) - h(y)\|^2 = 0 \iff \|h(x + y) - h(x) - h(y)\| = 0.$$

Since $\|\cdot\|$ is a norm, we conclude that $h(x + y) = h(x) + h(y)$. Similarly, let $x \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$. Then

$$\begin{aligned}
 \|h(\alpha x) - \alpha h(x)\|^2 &= \langle h(\alpha x) - \alpha h(x), h(\alpha x) - \alpha h(x) \rangle \\
 &= \langle h(\alpha x), h(\alpha x) \rangle - \langle h(\alpha x), \alpha h(x) \rangle - (\langle \alpha h(x), h(\alpha x) \rangle - \langle \alpha h(x), \alpha h(x) \rangle) \\
 &= \langle h(\alpha x), h(\alpha x) \rangle - \alpha \langle h(\alpha x), h(x) \rangle - \alpha \langle h(x), h(\alpha x) \rangle + \alpha^2 \langle h(x), h(x) \rangle \\
 &= \langle \alpha x, \alpha x \rangle - \alpha \langle \alpha x, x \rangle - \alpha \langle x, \alpha x \rangle + \alpha^2 \langle x, x \rangle \\
 &= 2\alpha^2 \langle x, x \rangle - 2\alpha^2 \langle x, x \rangle \\
 &= 0.
 \end{aligned}$$

Then we have

$$\|h(\alpha x) - \alpha h(x)\|^2 = 0 \iff \|h(\alpha x) - \alpha h(x)\| = 0.$$

Hence, $h(\alpha x) = \alpha h(x)$. Thus, (1) and (2) imply that h is a linear transformation. ■